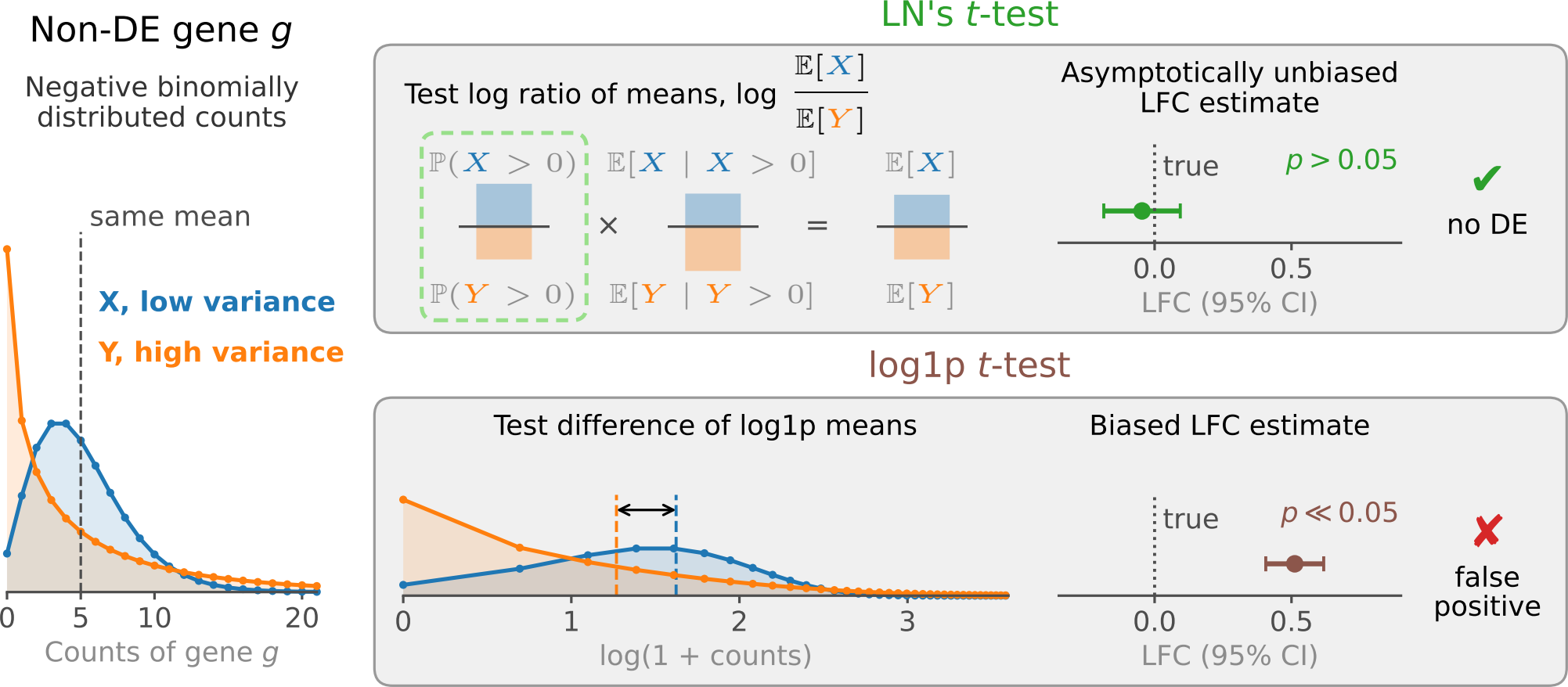

Non-DE gene g
Negative binomially distributed counts
same mean
X, low variance
Y, high variance
0
5
10
20
Counts of gene g
LN's t-test
Test log ratio of means, $\log \frac{\mathbb{E}[X]}{\mathbb{E}[Y]}$
$\mathbb{P}(X > 0)$
$\mathbb{E}[X \mid X > 0]$
$\mathbb{E}[X]$
$\mathbb{P}(Y > 0)$
$\mathbb{E}[Y \mid Y > 0]$
$\mathbb{E}[Y]$
Asymptotically unbiased LFC estimate
true
$p > 0.05$
0.0
0.5
LFC (95% CI)
no DE
log1p t-test
Test difference of log1p means
0
1
2
3
log(1 + counts)
Biased LFC estimate
true
$p \ll 0.05$
0.0
0.5
LFC (95% CI)
false positive